

Robustness checks

Table ESM 2. Results of robustness checks: Top 5% and Top 1% quality-weighted productivity models

term	Top 5% (journal prestige-weighted productivity)		Top 1% (Journal prestige-weighted productivity)	
	estimate	p.value	estimate	p.value
(Intercept)	-9.859	<0.001	-12.158	<0.001
acage_c	0.010	0.001	0.018	0.001
Discipline: BIO	-2.914	<0.001	-3.221	<0.001
Discipline: CHEM	-8.165	<0.001	-8.858	<0.001
Discipline: CHEMENG	6.676	<0.001	5.064	<0.001
Discipline: COMP	7.057	<0.001	4.181	<0.001
Discipline: EARTH	3.800	<0.001	3.946	<0.001
Discipline: ENER	8.234	<0.001	7.051	<0.001
Discipline: ENG	6.207	<0.001	4.835	<0.001
Discipline: ENVI	3.813	<0.001	1.981	<0.001
Discipline: MATER	-3.191	<0.001	-3.163	<0.001
Discipline: MATH	2.707	<0.001	3.004	<0.001
Discipline: MED	-2.082	<0.001	-3.577	<0.001
Discipline: NEURO	-1.653	<0.001	-0.839	0.064
Discipline: PHARM	-4.739	<0.001	-4.992	<0.001
Discipline: PHYS	-8.927	<0.001	-9.182	<0.001
ATS_c	-0.004	0.509	-0.002	0.863
coop_c	0.793	0.001	0.802	0.121
coopint_c	0.163	0.125	-0.062	0.784
inst_typeRest	-0.045	0.465	-0.042	0.715
ln_pubs_c	8.732	<0.001	4.653	<0.001
ln_pubs_c:mean_journal_pct_c	0.166	<0.001	0.131	<0.001
ln_pubs2_c	3.676	<0.001	3.040	<0.001
MaleMale	0.076	0.195	-0.048	0.682
mean_journal_pct_c	0.020	0.013	-0.041	0.009
Period1998-2003	-3.604	<0.001	-3.679	<0.001
Period2004-2009	-6.664	<0.001	-7.433	<0.001
Period2010-2015	-10.590	<0.001	-10.483	<0.001
Period2016-2021	-17.091	<0.001	-16.537	<0.001
prestmed_c	0.026	<0.001	0.032	<0.001

Table ESM 3. Robustness checks using population-averaged logit models (GEE)

term	Top 20% FC_PW (population-averaged logit, GEE)		Top 10% FC_PW (population-averaged logit, GEE)	
	estimate	p.value	estimate	p.value
(Intercept)	-2.580	<0.001	-6.657	<0.001
acage c	0.007	<0.001	0.010	<0.001
Discipline: BIO	-3.125	<0.001	-2.809	<0.001
Discipline: CHEM	-6.235	<0.001	-7.212	<0.001
Discipline: CHEMENG	6.783	<0.001	6.648	<0.001
Discipline: COMP	6.939	<0.001	7.074	<0.001
Discipline: EARTH	3.622	<0.001	3.780	<0.001
Discipline: ENER	8.359	<0.001	8.075	<0.001
Discipline: ENG	5.613	<0.001	6.130	<0.001
Discipline: ENVI	3.591	<0.001	3.849	<0.001
Discipline: MATER	-2.491	<0.001	-2.600	<0.001
Discipline: MATH	1.806	<0.001	2.219	<0.001
Discipline: MED	-0.068	0.432	-1.219	<0.001
Discipline: NEURO	-2.272	<0.001	-1.989	<0.001
Discipline: PHARM	1.377	<0.001	-1.842	<0.001
Discipline: PHYS	-8.021	<0.001	-8.563	<0.001
ATS c	0.003	0.390	-0.008	0.053
coop c	0.112	0.336	0.511	0.004
coopint c	-0.105	0.065	0.086	0.267
inst typeRest	0.028	0.465	-0.137	0.003
ln pubs c	14.013	<0.001	10.834	<0.001
ln pubs c:mean journal pct c	0.205	<0.001	0.185	<0.001
ln pubs2 c	4.238	<0.001	3.765	<0.001
MaleMale	0.041	0.256	0.012	0.787
mean journal_pct c	0.144	<0.001	0.064	<0.001
Period1998-2003	-2.535	<0.001	-3.151	<0.001
Period2004-2009	-4.851	<0.001	-5.907	<0.001
Period2010-2015	-8.344	<0.001	-9.777	<0.001
Period2016-2021	-14.709	<0.001	-16.156	<0.001
prestmed c	0.006	0.030	0.017	<0.001

Table ESM 4. Robustness checks using spline specifications for productivity

term	Top 20% (quality-weighted productivity, spline)		Top 10% (quality-weighted productivity, spline)	
	Estimate	p.value	Estimate	p.value
(Intercept)	-26.111	0.000	-203.165	0.000
acage c	0.007	0.000	0.010	0.000
Discipline: BIO	-3.070	0.000	-2.699	0.000
Discipline: CHEM	-6.114	0.000	-6.970	0.000
Discipline: CHEMENG	7.090	0.000	6.863	0.000
Discipline: COMP	6.964	0.000	7.005	0.000
Discipline: EARTH	3.617	0.000	3.732	0.000
Discipline: ENER	8.512	0.000	8.000	0.000
Discipline: ENG	5.707	0.000	6.189	0.000
Discipline: ENVI	3.555	0.000	3.739	0.000
Discipline: MATER	-2.526	0.000	-2.605	0.000
Discipline: MATH	1.767	0.000	2.189	0.000
Discipline: MED	-0.070	0.264	-1.150	0.000
Discipline: NEURO	-2.210	0.000	-1.885	0.000
Discipline: PHARM	1.239	0.000	-1.731	0.000
Discipline: PHYS	-7.887	0.000	-8.324	0.000
ATS c	0.005	0.208	-0.006	0.179
coop c	0.093	0.385	0.439	0.006
coopint c	-0.110	0.053	0.078	0.312
inst typeRest	0.038	0.319	-0.153	0.001
MaleMale	0.041	0.253	0.001	0.979
mean journal pct c	0.187	0.000	0.087	0.000
ns(ln_pubs_c, df = 3)1	26.715	0.000	130.476	0.000
ns(ln_pubs_c, df = 3)2	116.036	0.000	418.678	0.000
ns(ln_pubs_c, df = 3)3	143.167	0.000	215.395	0.000
Period1998-2003	-2.574	0.000	-3.258	0.000
Period2004-2009	-4.874	0.000	-5.969	0.000
Period2010-2015	-8.325	0.000	-9.749	0.000
Period2016-2021	-14.573	0.000	-15.942	0.000
prestmed c	0.006	0.014	0.017	0.000